

INDIANAPOLIS EAGLE CREEK, IN, USA

WMO: 724384

Lat: 39.825N

Lon: 86.296W

Elev: 251

StdP: 98.35

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53842

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -16.3	(c) -13.2	(d) -21.1	(e) 0.6	(f) -15.2	(g) -18.5	(h) 0.8	(i) -12.5	(j) 10.8	(k) -3.0	(l) 9.6	(m) -2.5	(n) 3.1	(o) 300	(p) 0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.4	(c) 32.6	(d) 23.5	(e) 31.5	(f) 23.0	(g) 30.3	(h) 22.3	(i) 25.2	(j) 30.5	(k) 24.4	(l) 29.3	(m) 23.7	(n) 28.3	(o) 3.3	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 23.5	(b) 18.9	(c) 28.2	(d) 22.8	(e) 18.0	(f) 27.4	(g) 22.2	(h) 17.4	(i) 26.9	(j) 78.0	(k) 30.5	(l) 75.0	(m) 29.4	(n) 71.9	(o) 28.3	(p) 28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.9	(b) 7.9	(c) 7.1	(e) -19.8	(f) 34.1	(g) 3.6	(h) 1.7	(i) -22.4	(j) 35.3	(k) -24.5	(l) 36.3	(m) -26.5	(n) 37.2	(o) -29.2	(p) 38.4		
(a) 8.9	(b) 7.9	(c) 7.1	(e) -20.2	(f) 26.3	(g) 3.6	(h) 1.0	(i) -22.7	(j) 27.0	(k) -24.8	(l) 27.6	(m) -26.8	(n) 28.1	(o) -29.4	(p) 28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-2.0	-0.2	5.9	12.3	17.8	22.5	24.2	23.5	19.9	13.1	6.4	0.6
	DBStd	10.54	6.64	6.38	6.17	5.16	4.52	3.23	2.96	2.70	3.94	4.96	5.28	5.90
	HDD10.0	1316	376	291	159	34	2	0	0	0	0	24	132	297
	HDD18.3	2915	632	519	387	189	67	7	1	1	28	176	358	551
	CDD10.0	2070	3	5	32	104	244	376	441	419	298	119	25	4
	CDD18.3	627	0	0	2	9	50	133	183	162	77	13	0	0
	CDH23.3	5295	0	0	11	76	411	1088	1666	1313	633	96	1	0
	CDH26.7	1686	0	0	1	7	96	349	598	414	199	21	0	0
Wind	WSAvg	3.0	3.7	3.6	3.6	3.7	2.9	2.6	2.4	2.2	2.3	2.8	3.2	3.4
Precipitation	PrecAvg	1120	75	66	88	114	119	132	109	82	86	79	86	84
	PrecMax	1401	243	151	163	211	242	275	287	169	230	201	202	202
	PrecMin	848	22	25	22	45	45	0	24	9	10	23	15	29
	PrecStd	168	46	34	38	44	49	56	68	37	55	43	42	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	19.2	25.0	27.7	31.2	32.9	35.2	33.6	32.9	29.3	22.3	17.4
		MCWB	14.2	13.4	16.7	17.8	21.3	23.2	24.9	23.7	22.0	20.8	15.2	13.7
	2%	DB	13.1	15.8	21.6	25.9	29.0	31.6	32.7	32.0	31.2	26.1	19.3	14.1
		MCWB	10.7	12.3	14.9	16.8	20.1	22.5	24.4	23.4	22.0	18.9	14.2	12.0
	5%	DB	10.6	12.4	18.3	23.5	27.6	30.3	31.4	30.5	29.0	23.5	17.3	11.7
		MCWB	8.4	9.4	13.0	15.7	19.4	22.0	23.6	22.7	21.0	17.5	13.2	9.7
	10%	DB	7.5	9.3	16.0	21.3	25.9	28.8	30.0	28.9	27.1	21.5	14.8	8.9
		MCWB	5.2	6.1	11.7	14.9	19.1	21.1	22.9	22.0	19.9	16.6	11.0	6.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	15.2	17.5	20.0	22.9	25.0	26.6	25.7	24.3	22.2	17.4	14.8
		MCDB	16.9	18.0	23.4	25.7	28.0	30.9	32.0	31.4	29.6	26.8	20.9	16.7
	2%	WB	11.6	12.6	15.8	18.4	21.7	24.0	25.6	24.7	23.2	20.6	15.4	12.3
		MCDB	12.7	14.6	20.3	22.9	27.1	29.5	30.8	29.8	28.4	24.4	18.1	13.8
	5%	WB	8.7	9.9	14.3	17.0	20.8	23.2	24.9	24.0	22.3	19.0	13.7	9.7
		MCDB	10.2	12.9	17.5	21.4	25.7	28.5	29.8	28.5	26.9	22.3	16.6	11.4
	10%	WB	5.3	6.3	12.0	15.6	19.8	22.3	24.0	23.2	21.3	17.2	11.5	7.1
		MCDB	7.7	9.2	15.5	20.1	24.4	27.1	28.5	27.3	25.3	20.3	14.5	8.6
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.6	9.8	10.7	10.2	9.7	9.4	9.6	10.6	10.1	9.0	7.4
		MCDBR	10.9	12.3	13.0	13.2	12.2	11.1	10.3	10.6	12.6	12.6	12.0	10.1
		MCWBR	9.3	9.9	8.1	7.2	5.6	5.0	4.1	4.4	5.0	6.6	8.3	8.7
	5% WB	MCDBR	10.2	11.4	11.4	10.7	10.0	9.7	9.2	9.1	10.1	10.0	10.6	9.8
		MCWBR	9.5	10.1	7.4	7.0	5.3	4.9	4.3	4.5	4.6	5.8	8.6	9.0
Clear-Sky Solar Irradiance	taub	0.302	0.324	0.347	0.404	0.438	0.440	0.455	0.452	0.417	0.359	0.320	0.300	
	taud	2.433	2.351	2.369	2.219	2.174	2.208	2.168	2.163	2.245	2.407	2.465	2.456	
	Ebn at Noon	866	892	907	870	845	841	824	815	821	841	837	839	
	Edh at Noon	81	101	111	137	147	142	147	144	124	95	78	74	
All-Sky Solar Radiation	RadAvg	1.75	2.58	3.64	4.74	5.41	6.16	6.02	5.48	4.52	3.16	2.10	1.47	
	RadStd	0.18	0.29	0.31	0.39	0.49	0.38	0.35	0.32	0.37	0.30	0.23	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+107	N/A

Nomenclature: See separate page